



Do Habitat Edges Really Concentrate Bat Activity?

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Introduction

- Wisconsin bat habitat selection is tricky to study
- Examining bat habitat selection will create a better understanding of optimal detector placement (interior, edge, or open areas).
- Aim to create an acoustic data set to study the magnitude of bat activity.

RQ: Does habitat placement of detectors influence recorded bat call frequency after accounting for recording effort and seasonal timing?

H₀: Habitat location has no effect on bat activity

H₁: Location significantly affects bat activity

Methods

- Study location in Schmeckle Reserve in Stevens Point, WI.
- Conducted from Oct. 4th to Nov. 1st, with maintenance visits every week.
- Deployed 9 AudioMoth detectors, dividing them between 3 sites with an interior, edge, and open habitat detector.
- Bat calls were analyzed with Kaleidoscope Pro and the Calls per night data was modeled with an NB GAMM:
 - **Calls per Night** ~ **Habitat** + s(**Night**) + random **Site** + random **Detector** + offset(log(**Recording hours**))



Fig 1. Photos of AudioMoth detectors at the **open** (left), **edge** (center), and **interior** (right) at the MC site.



Fig 2. Map of detector locations in Schmeckle Reserve. Survey locations are Sedge Meadow (SM), Lake Prairie (LP), and Moses Creek Wetland (MC).

Results

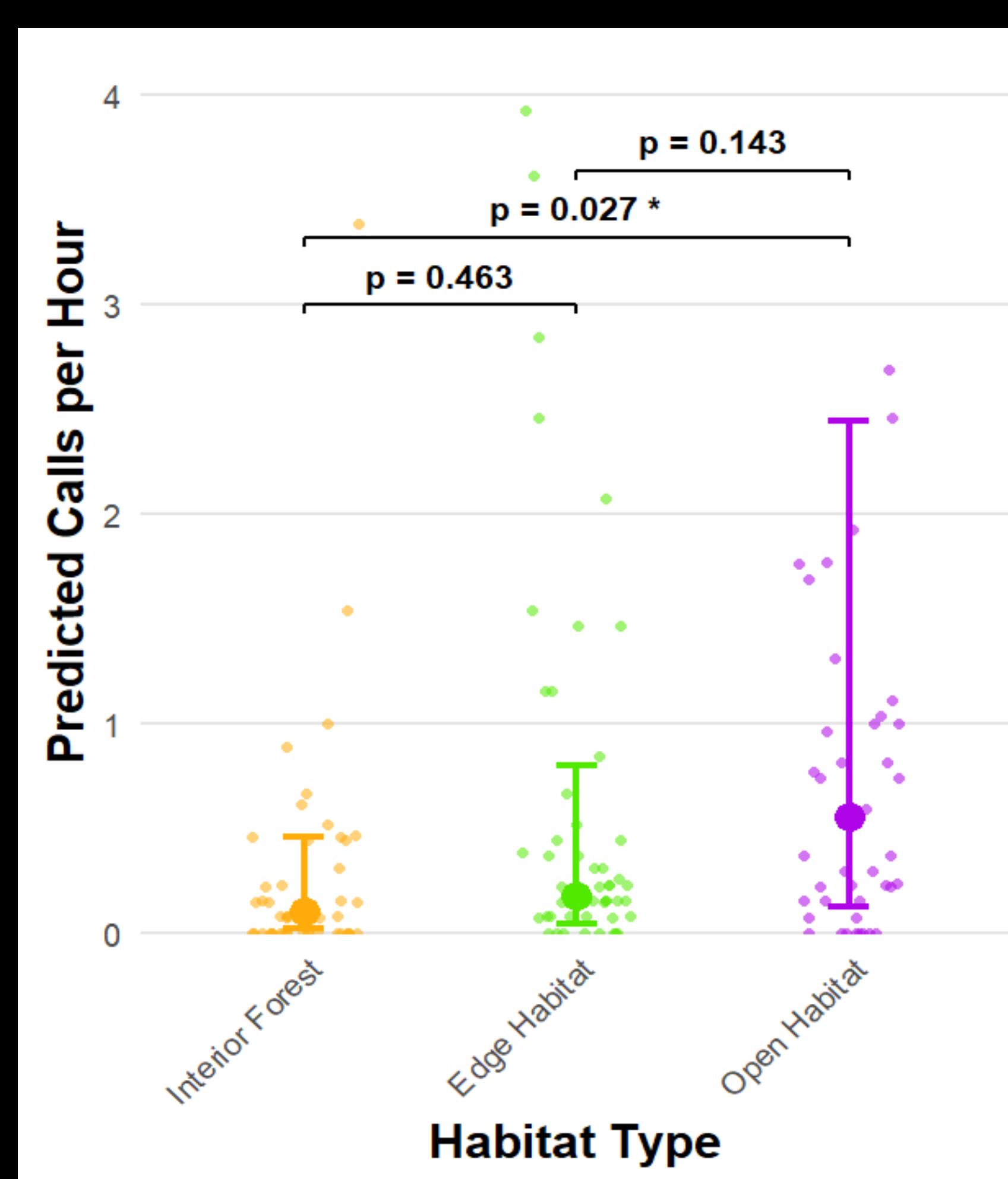


Fig 3. Habitat Effects. Predicted bat activity (calls/hour) by habitat. Error bars are a 95% CI and P values are a pairwise comparison. **Open** (n=65 nights; $\bar{x}$ =0.552) had significantly more calls than **interior** (n = 49; $\bar{x}$ =0.103). **Edge** (n = 61; $\bar{x}$ =0.180) had no significant differences.

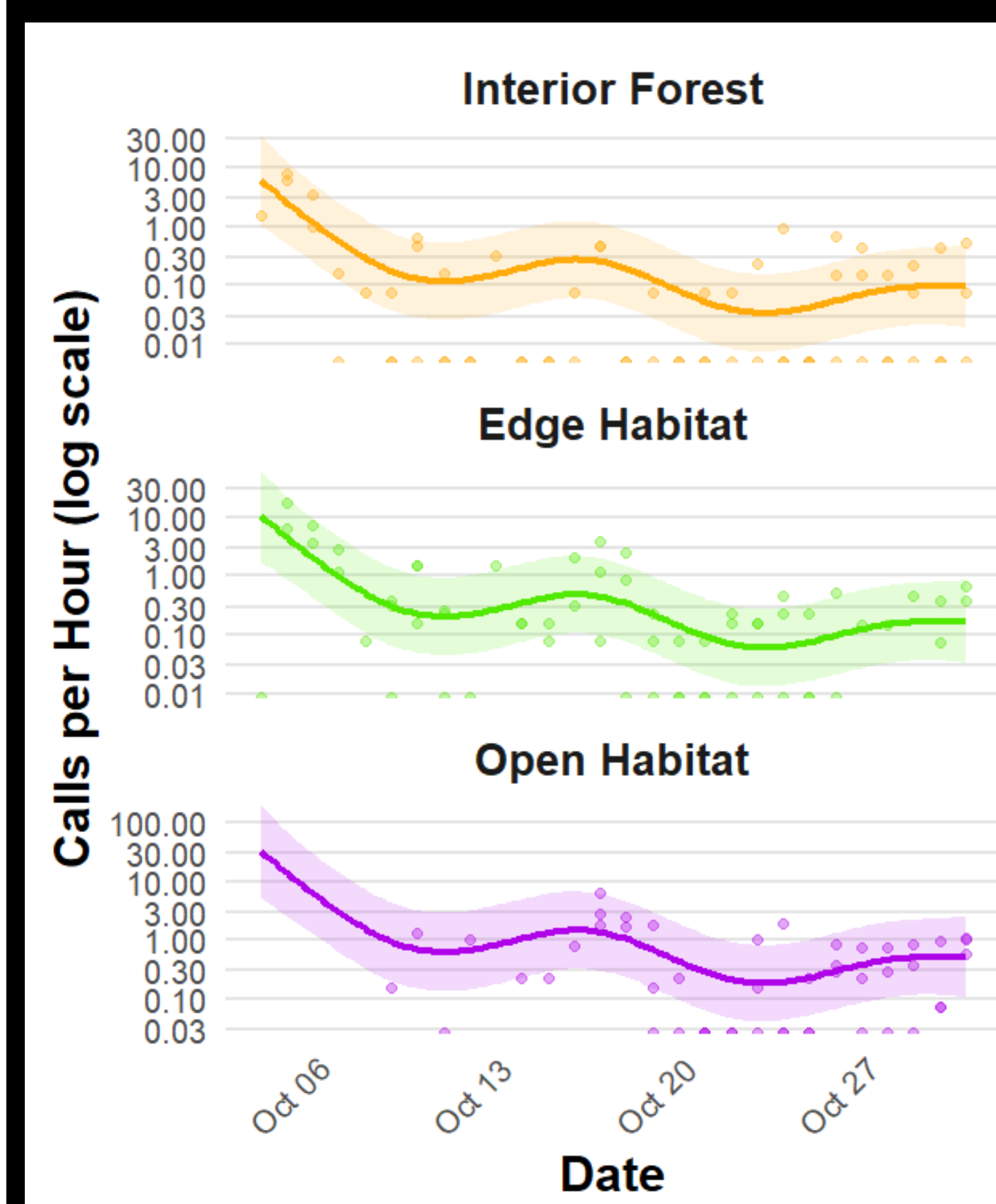


Fig 4. Raw Nightly Bat Activity by Habitat. Points are observed calls per night, and lines show population-level NB GAMM predictions from with seasonal smoothing and effort adjustment on a log scale. Trend line accounts for all **model parameters**.

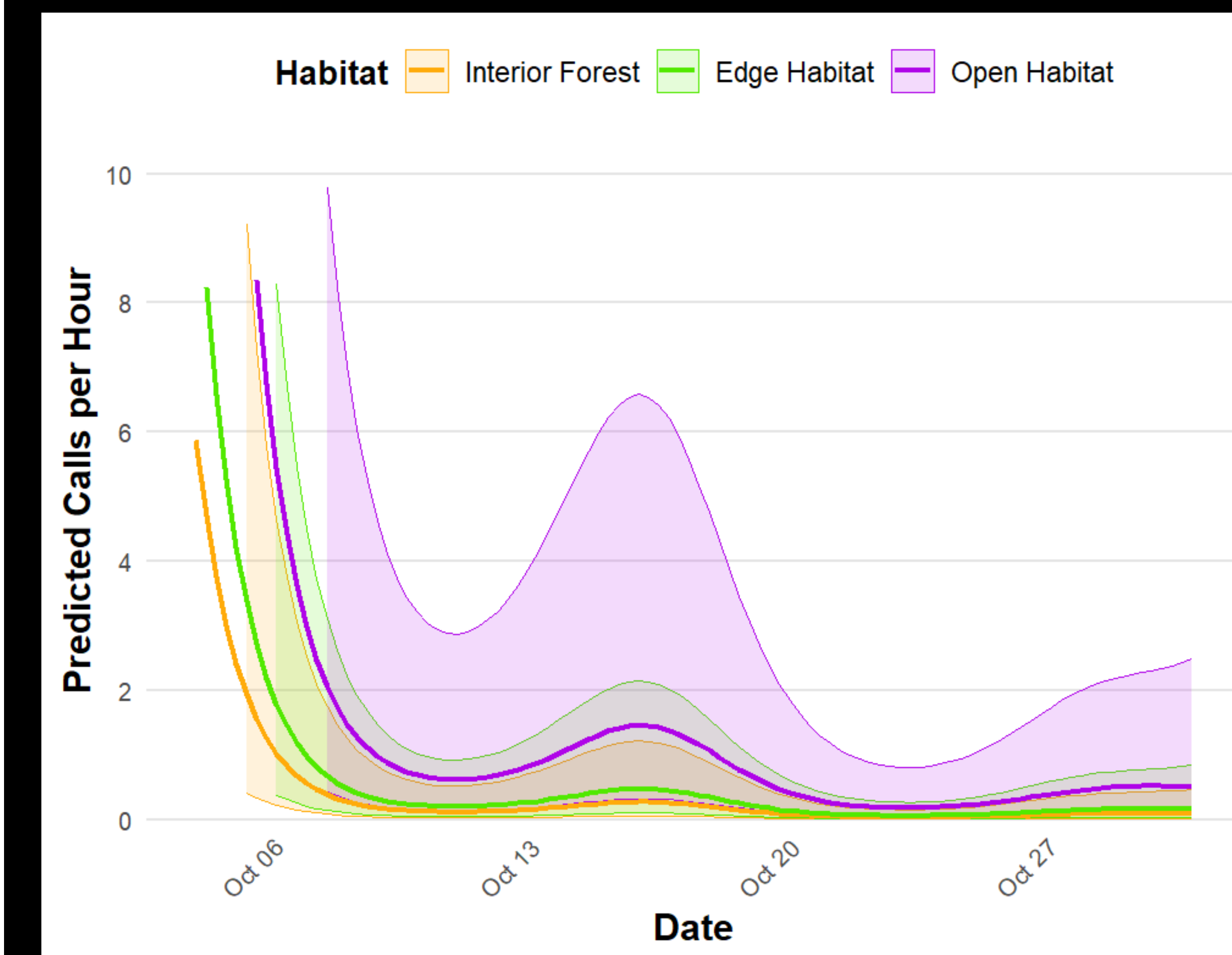


Fig 5. Seasonal trends. Bat activity by habitat predicted by NB GAMM. Shaded areas = 95% CI. A large **migration pulse** happened Oct 06-07, warranting a seasonal smooth.

Discussion

- Bat call activity was found to be highest in open habitats compared to interior and edge forest habitats.
 - AudioMoths should be placed in open areas for optimal detection.
- Open areas had slightly more activity than edge detectors, but there was no significant difference found.
 - Bats in open areas could also utilize edge habitats.
- The migration could have influenced results.
 - Migrating bats might select for open habitat.
 - Future study could be done at a different time of year.

Acknowledgements

We would like to thank Schmeckle Reserve for allowing us to put up our detectors.

Supplemental

